

Performance Testing

Date	15 March 2026
Team ID	
Project Name	Smart Lender AI – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	User Login, User Registration, Loan Prediction, Dashboard
Test Environment	Render Cloud Deployment (Flask + SQLite)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	20	60sec	Successful login without errors
2	User Registration	15	60sec	New account created successfully
3	Loan Prediction	30	120sec	Prediction generated quickly

4	Dashboard Access	25	60sec	Dashboard loads without delay
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.10 sec	Pass	Within limit
3	Throughput (Req/sec)	-	45 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	42%	Pass	Normal usage
6	Memory Utilization	< 80%	51%	Pass	Memory usage acceptable

Step 4: Observations & Analysis

Key Findings

The Smart Lender AI application handled multiple user requests efficiently. Login, registration, dashboard loading, and loan prediction completed successfully with low response times. No request failures occurred during testing.

Bottlenecks Identified

No significant bottlenecks were observed. A slight increase in response time occurred during simultaneous prediction requests but remained within acceptable limits.

Optimization Steps Taken

- Optimized Flask routing.
- Reduced unnecessary database queries.
- Loaded the trained ML model only once during application startup.
- Optimized static files (CSS, images).
- Improved feature preprocessing for faster prediction.

Step 5: Screenshots / Evidence



